

Cybersecurity Plan For An Author's Small Business

Businesses of all sizes confront common threats in cyberspace, including viruses, malware, phishing, and ransomware. Cybersecurity is essential because in 2004, 3,200 U.S. data breaches impacted 350 million people (Kelly, 2024). Attacks destroy trust, and companies often lose both customers and their reputations. A business might also face fines and lawsuits if customer data is stolen.

Attackers use many tools and methods, and even companies with cybersecurity teams have experienced breaches. Small businesses are especially at risk. Many owners don't know enough to understand and prevent attacks, and human error often causes breaches. In 2022, 43% of ransomware attacks targeted small businesses (Wetzig, 2022).

A cybersecurity strategy defends against these attacks and provides a plan to use if an attack happens. Steps must be taken to protect physical computers, online accounts, websites, and data.

Focus Areas

A small business should focus on three areas:

- **Technologies:** Use antivirus software and password managers.
- **Processes:** Create clear cybersecurity rules and train everyone (including subcontractors).
- **Access Controls:** Divide business information, giving access only to those who need it, including contractors like cybersecurity experts, editors, and proofreaders. Use two-factor authentication for all users (Cesaire, n.d.).

Implementation Areas

It's crucial to protect customer data, financial information, and sales history. Key steps include:

- Define and separate roles and responsibilities
- Enable multifactor authentication (MFA)
- Implement firewalls
- Encrypt data, both of physical computers and online
- Establish additional security measures

The Business: Overview

This cybersecurity plan is for a small author website. The website stores email addresses but doesn't keep financial data, using external payment systems instead. Currently, the website only links to book retailers, but the author plans to sell writing courses through apps like PayPal and Venmo, and use drop shippers for merchandise. The author has a blog and links to social media on the website. The author also has copyrighted materials and newsletters on the site.

The Business: Copyrighted Material

Reader magnets, such as free stories or character sheets, are copyrighted materials offered to engage readers, which are under copyright. Readers who sign up for the newsletter can opt out anytime using the unsubscribe button at the email's footer. Some authors put their works on their websites with links to hidden pages for editors, proofreaders, and others to use. This data must also be protected. People who like the books can sign up to be on the beta reader team (who read and critique the book while it's being written) and the alpha reader team (who want an early copy).

Thieves steal authors' works to publish and sell them, so it's important to protect these materials. It's also important to prevent viruses when sharing files, sending emails, and downloading materials. Ransomware can lock up important data, which can cause financial problems. Protecting email lists and limiting the collection of other data is critical.

The website

The website was built using a simple, free, one-page template altered to fit the business. Terraform was used to set up the:

- Website and data storage, including newsletters and reader magnets (with plans to add entire works in the future).
- A table for newsletter signups, allowing users to receive updates, news, and giveaways.
- Tables for beta and alpha reader team signups (planned for later).
- A key vault to store sensitive items like keys and SSL certificates.
- A firewall that only allows HTTPS websites.

Goals

To address these threats, the goals include:

- Develop and implement cloud and local infrastructure to protect against cyber threats.
- Draft and document an incident response plan.
- Write, test, and deploy the infrastructure.

Write, Document, and Implement the infrastructure

The infrastructure is written in Terraform, which uses code to set things up. Terraform, a scripting language, makes it easier to set up accounts, databases, tables, and more with code. VSCode, a free editor, works with platforms like Azure, AWS, and Google Cloud, making development simpler.

The infrastructure was written in VSCode, checked for errors, and then run. Problems happened after several months because of code changes, which could have been avoided by specifying a Terraform version. However, using the latest code is usually better. The entire infrastructure, including firewalls, the storage account for the website (newsletters and emails), the email signup table, and a key vault, was created, tested, and run to build the website.

Firewalls

The firewalls in Terraform only allow HTTPS (so users only use secure websites). Website users don't log in. They can only see the landing page. They can click on a newsletter to read it; newsletters are stored on the website. There's also a button to email the author, which opens the user's email program with the author's Gmail address. There are social media links at the bottom if the user wants to read the author's posts.

Users can sign up with their email address to get newsletters, reader magnets, and updates. Clicking 'Sign Up' sends a welcome email with download buttons for reader magnets. Reader magnets are stored securely on the website. Clicking the download button in the email downloads the reader magnet.

Email addresses are stored in a secure table without additional data that could enable identity theft or financial fraud (Dawson, 2025). The author does not share or sell email addresses. Since financial and personal identifying information is carefully regulated, third-party vendors, such as drop shippers, should be used so this data is never seen by the author or saved on the website.

Email addresses are stored in a secure table without data that could enable identity theft (Dawson, 2025). The author doesn't share or sell email addresses. Because financial and personal information is carefully regulated, the author should use payment services and drop shippers to avoid handling this data. Users can unsubscribe in the email, which deletes their information from the table.

The author could also periodically send "Do you still want to hear from me" emails to delete them from the database if desired. Email users have the right to be forgotten, which is the erasure of data (in this case, an email address) if the patron wants it deleted, immediately upon request (pressing the Unsubscribe button) (Wolford, n.d.). An event was also created to delete any invalid email after a group email is sent to subscribers.

Refined Critical Tasks:

- **Create a system to delete email addresses:** Develop a system that automatically removes an email address when someone clicks the unsubscribe button.
- **Add an unsubscribe button to emails:** Put an unsubscribe button in every email.
- **Link the button to the system:** Make sure each unsubscribe button works with the system to delete the email.
- **Testing the systems:**
 - Sign up for a newsletter
 - Download all three formats
 - Unsubscribe and check that the email was deleted

Role-Based Access Control (RBAC)

Setting up roles is key to cybersecurity. By giving each role only the necessary permissions, you limit damage if there's a breach and protect important items like encryption keys and the author's works. Logging in as the root user, with all permissions, is very risky.

For this project, a Terraform Admin role was created with limited permissions to set up resources like storage accounts and key vaults. Access to sensitive items, like key vault contents, is limited. A Security Admin role was also created to monitor logs and handle security threats (The future is yours, 2025).

To ensure secure implementation:

- Create the first storage directly through the cloud provider to set up roles.
- Limit Terraform Admin permissions to creating and deleting resources.
- Use shared access signatures (SAS) for temporary access during future changes, with expiration dates to revoke access (Cesaire, n.d.).

If financial or complex user data is stored later, the author will follow financial regulations. (Kelly, 2024). For now, using payment services and shipping companies means no private data, besides email addresses, is kept.

Passwords

A strong password policy is important. This includes rules like:

- Passwords must be at least 10 characters long.
- Passwords must have uppercase and lowercase letters, at least one number, and a symbol (like ! or *).
- Passwords should be changed every 60 to 90 days and not reused.

Microsoft Azure creates and stores passwords for its users. It's important to protect these complex passwords. A password manager (like Microsoft Edge, Proton Pass, or NordPass) adds security by storing and creating strong passwords. The author should set reminders for users to change passwords every 60-90 days.

Multi-Factor Authentication (MFA)

Multi-Factor Authentication (MFA) adds security by requiring two or more ways to prove your identity. These are:

- Something you know (like a password).
- Something you have (like a smartphone).
- Something you are (like a fingerprint) (Multifactor Authentication, 2023).

Providers like Azure often have users download an authenticator app for their phone. Both the Terraform Admin and Security Admin users installed an authenticator app, which protects the account. New users, like editors, must use MFA to log in and download content.

Encryption: Website and Computer

Encryption is important for the website, the author's work, and the author's computer. The author uses Google Docs to store works. MFA is required for Google Drive, and shared access is also protected by MFA. Google Workspace encrypts files so even Google can't read them. Also, Docs, Sheets, and Slides are encrypted by default (Get started with encrypted files in Drive, Docs, Sheets & Slides, n.d.)

Works on the website are in Azure Blob, which also encrypts data. When something is downloaded from Azure Blob, it uses a \$web container to secure the data (Download static website contents, n.d.). The "Secure transfer" option can be enabled to make sure any requests made must take place over secure connections. A malicious actor using HTTP will fail (Security recommendations for Blob storage, 2025).

Storing works on Google Drive and Azure Blob helps prevent ransomware. Encrypting data and backing up Blob data in different zones reduces the impact of attacks. Firewalls and encryption protect the user emails in the blob's table. So, a ransomware attack wouldn't do much damage.

Windows 11 typically has device encryption enabled by default. To check or turn it on, log in as Administrator, open 'Settings' via the Start button, navigate to 'Privacy and Security,' and turn 'Device Encryption' on (Morton, 2024). This prevents ransomware attacks on the computer.

Authors who write in public (like coffee shops) should use hardware encryption and a VPN, and turn off automatic connections. They should never use public Wi-Fi without a VPN and protect their work with Google Docs.

Antivirus software and Windows Security Check add protection. Encrypted storage keeps data safe even if the computer is lost. OneDrive offers backups, but Google Drive and Azure Blob may be enough.

Other Security Measures

Microsoft Defender for Cloud is enabled across all accounts, with specific settings for website storage. Other security measures include:

- Use resource locks on the storage account to prevent accidental deletion
- Turn on soft delete for blobs, to allow data recovery after deletion
- Create a Key Vault to manage encryption keys with Terraform
- Only allow HTTPS for all data transfers, including reader magnets and newsletters, through firewall rules

- Disallow cross-tenant object replication (so someone using another account can't copy the data)
- Only allow TLS 1.2 (or higher) for secure connections
- Check the blob container logs daily and automatically delete them after 90 days (set up in Terraform)
- Set up Azure Monitor alerts for Blob storage security advice (Security recommendations for Blob storage, 2025)

Employee Training

All users must be trained on MFA, role-based access control, phishing awareness, and how to handle ransomware and impersonation attacks. Contractors, like editors, must use secure communication. Check all unusual requests, like requests to send money, by talking to the person directly (by phone or video call). Change logins and keys regularly.

Document the breach plan

A Cybersecurity Incident Response Plan (CSIRP) explains what to do if there's a security breach. Many groups require reporting breaches within 72 hours, but the rules vary. California requires a CSIRP (Kelly, 2024). Microsoft and Google have their own plans, but this CSIRP is for the author's specific systems.

- **Prepare:** Put all security measures in place before anything happens. These include firewalls, MFA, role-based access controls, encryption, backups, and user training.
- **Detect:** Set up alerts for suspicious activity, like many failed logins. Use antivirus software to find infections.
- **Act:** If there's a breach, act right away. This might mean shutting down infected computers, looking for what happened, restoring data from backups, and changing all passwords.
- **Notification:** Notify people right away if their data is stolen, even if you don't have to by law. Honesty engenders trust. Use payment services and drop shippers to avoid handling customer financial details or personal information.

Implementation of Plans

The first storage accounts and Terraform admin roles were set up before Terraform was used. Then, firewalls, storage, and the Key Vault were created with Terraform. After that, Microsoft Defender was turned on, an SSL certificate was obtained, the author was trained, the computer was encrypted, and logs were monitored.

Final Review

Following thorough review and verification, including user training and system testing, the implemented security measures have proven effective. The system exhibits robust security posture, with no breaches recorded to date.

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